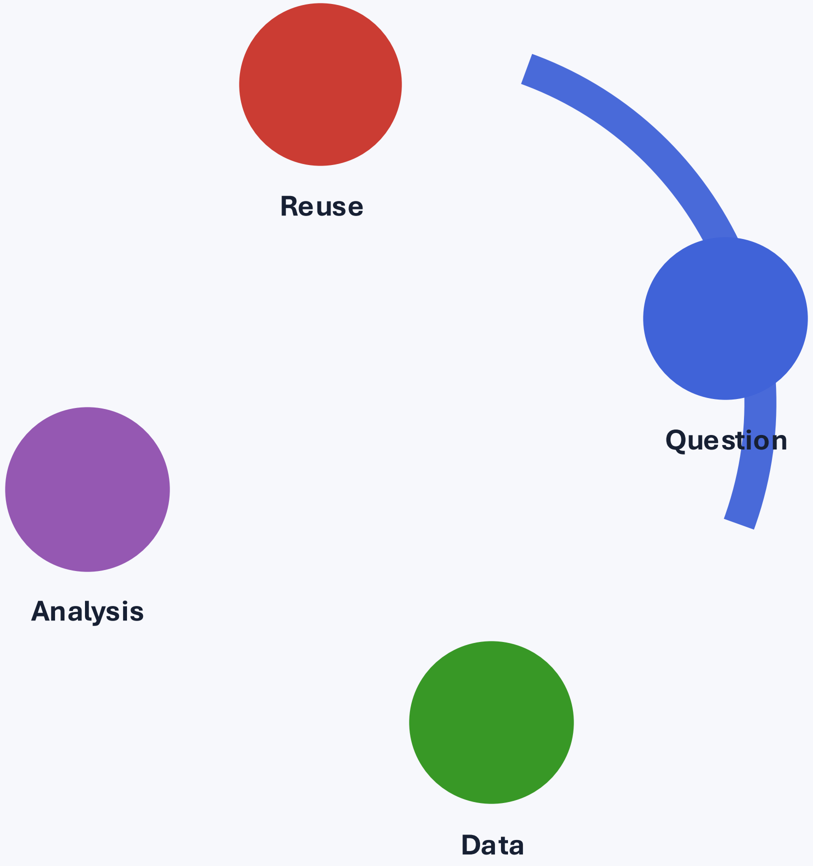
From a Research Question to Reusable Data

An introduction to biomedical data science, the data lifecycle, and FAIR principles



BIOE 860 • Week 1

Today’s question

What has to happen between a good scientific question and a trustworthy result?

A result is only the end of a chain.

Question → **Collect** → **Organize** → **Analyze** → **Share / Reuse**

At every step, choices affect what can be learned later.

A fictional biomedical study

We will use one small study throughout today's class.

Research question

Among adults in a metabolic-health study, how are BMI and fasting glucose related to diabetes status?

Data collected

Participant ID, age, sex, visit date, height, weight, fasting glucose, and diagnosis.

Goal

Produce an analysis that another researcher could understand and reproduce.

Sounds simple. Then the spreadsheet arrives...

The spreadsheet arrives

Before calculating anything: what worries you?

patient_id	age	sex	visit_date	height	weight	glucose	diagnosis
P001	54	F	2026-01-05	165	74	102	No diabetes
P002	61	female	1/7/26	1.78 m	91	148	T2D
P003	-3	M	07-Jan-2026	180	83	NA	none
P003	47	Male	2026/01/07	180	83	96	No
P004	39	F	2026-13-01	172	72 kg	520	type 2
P005	67	Woman		160	58	110	0

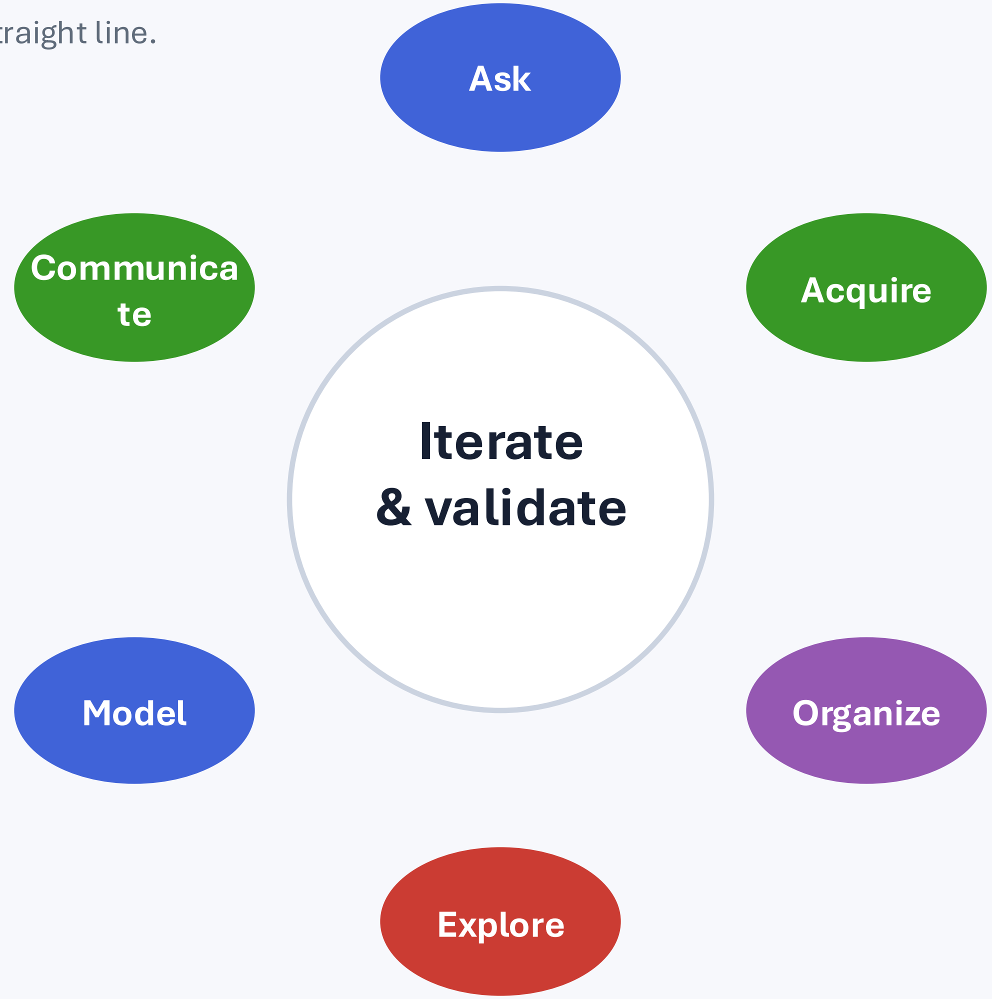
Take 2 minutes. Mark every problem you can find — and explain why it matters.

What is “messy” about it?

- Values can be wrong: impossible ages, invalid dates, implausible measurements.
- The same concept can be coded many ways: F / female / Woman; T2D / type 2 / 1.
- Units can be mixed or undocumented: centimeters vs meters, kilograms embedded in cells.
- Identifiers and rows may not mean what we think: duplicate ID — duplicate person or repeat visit?
- Missingness can be ambiguous: blank, NA, 0, “unknown” may mean different things.
- Without documentation, even a clean-looking table can be unusable.

The data science lifecycle

Data science is an iterative process, not a straight line.



Good workflows make it possible to move backward as well as forward.

FAIR: a design goal for digital research objects

FAIR asks whether data, metadata, code, and workflows can be discovered and reused.

F	**Findable**	Can someone — or a computer — discover it?
A	**Accessible**	Once found, can it be retrieved under clear conditions?
I	**Interoperable**	Can it work with other data and systems?
R	**Reusable**	Is there enough context to use it correctly again?

FAIR ≠ "put everything on the internet."

F — Findable

Discovery starts with identifiers and metadata.

Persistent identifier

A stable identifier lets a dataset or workflow be referred to unambiguously over time.

Rich metadata

Title, creators, variables, population, dates, methods, keywords, and links help people and machines understand what exists.

Searchable resource

Registering or indexing the object makes discovery possible beyond the original lab.

Our messy file: “study_final_v7_REAL.xlsx” on one laptop

→ Hard to find, hard to identify, easy to lose.

A — Accessible

Access can be open, restricted, or controlled — but the process should be clear.

Retrievable

The identifier resolves to a place where the object can be requested or obtained.

Standard protocol

Access should use broadly implementable mechanisms rather than a one-off personal arrangement.

Authentication is allowed

FAIR explicitly permits authentication and authorization when needed.

Biomedical example

A controlled-access human dataset can be FAIR when metadata remain discoverable and the access procedure is documented.

I — Interoperable

Shared meaning matters as much as shared file formats.

Two files can both be CSV and still fail to interoperate.

Concept	Dataset A	Dataset B	Problem
Sex	F / M	female / male	Different vocabularies
Glucose	mg/dL	mmol/L	Different units
Diagnosis	T2D	E11.9	Local label vs standard code
Missing	NA	-999	Different missing-value conventions

Interoperability requires agreed structure, vocabularies, units, and relationships.

R — Reusable

Could a researcher five years from now use the data correctly?

- What exactly does each variable mean?
- How and when were measurements collected?
- What transformations or exclusions were applied?
- What are the units, allowable values, and missing-value rules?
- What license or data-use conditions apply?
- Where did the data come from, and what changed along the way? (provenance)

A data dictionary is not glamorous. It is often what makes reuse possible.

FAIR does not mean Open

These ideas overlap, but they answer different questions.

FAIR

Designed to support discovery, access, integration, and reuse by people and machines. Access may be restricted when justified.

Open

Generally emphasizes that an object can be accessed and used with minimal restriction. Openness alone does not guarantee good metadata or interoperability.

Possible: FAIR + controlled access

From messy to usable

Cleaning is only one part of the solution.

Problem	Action	What must be documented?
Mixed sex labels	Map to a defined coding scheme	Allowed values + mapping rule
Mixed height units	Convert to centimeters	Original units + conversion
Duplicate ID	Investigate row meaning	Participant vs visit identifier
Missing glucose	Use one missing representation	Reason for missingness if known
Diagnosis labels	Use defined vocabulary	Code system / category definitions

Never “fix” an ambiguous value by guessing.

Reproducibility: preserve the path, not just the answer

Raw data

Keep an unchanged source copy.

Code

Record every transformation and analysis.

Environment

Record software, packages, versions, and key settings.

Output

Generate tables/figures from code when possible.

Reproducible ≠ guaranteed correct. It means the computational path can be inspected and repeated.

Biomedical data add extra responsibilities

The “can we?” question is not the same as the “should we?” question.

Privacy & confidentiality

Identifiers, small cells, dates, and combinations of variables can create disclosure risk.

Consent & governance

Data use must be consistent with consent, agreements, institutional rules, and approved purposes.

Bias & representation

Who is represented, who is missing, and how measurements were obtained affect conclusions.

Good data science includes scientific, computational, and ethical judgment.

Activity: make the study more FAIR

You are taking over the project from another researcher. What would you ask for?

In 8–10 minutes, propose improvements in four categories:

Findable

How will the dataset and code be named, identified, described, and discovered?

Accessible

Where will they live? Who can retrieve them? Under what conditions?

Interoperable

What formats, units, variable names, and vocabularies should be standardized?

Reusable

What metadata, provenance, license/use conditions, and documentation are needed?

Bonus: Which improvements help reproducibility but are not explicitly captured by the acronym FAIR?

A practical project structure

Simple organization prevents many avoidable problems.

```
metabolic-study/
  README.md
  data/
    raw/
    processed/
  code/
    01_clean.R
    02_analyze.R
  metadata/
    data_dictionary.csv
  output/
    figures/
    tables/
```

Principles behind the structure

- Do not overwrite the raw data.
- Separate data, code, metadata, and outputs.
- Use meaningful names and a README.
- Make the analysis rerunnable.
- Track changes with version control.

What I want you to remember from Week 1

1. Start with the scientific question — not the software.
2. Inspect and understand data before analyzing them.
3. FAIR means Findable, Accessible, Interoperable, Reusable.
4. FAIR is compatible with controlled access.
5. Reproducibility depends on preserving data, code, context, and decisions.

Before next class

Short reading + a small reflection.

Required reading

Wilkinson MD et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. Scientific Data 3:160018.

Practical overview

GO FAIR: FAIR Principles. Use it as a quick reference for the individual F, A, I, and R sub-principles.

Reflection (5–10 minutes)

Choose a dataset you have used before. Identify one thing that made it easy to reuse and one thing that made it difficult. Map each observation to FAIR or reproducibility.

References & resources

Wilkinson MD, Dumontier M, Aalbersberg IJ, et al. The FAIR Guiding Principles for scientific data management and stewardship. Scientific Data. 2016;3:160018. doi:10.1038/sdata.2016.18

GO FAIR. FAIR Principles. https://www.go-fair.org/fair-principles/

Wilkinson MD, Sansone SA, Schultes E, et al. A design framework and exemplar metrics for FAIRness. Scientific Data. 2018;5:180118. doi:10.1038/sdata.2018.118

Teaching files included with this lecture: messy dataset, clean dataset, data dictionary, R script, student worksheet, and instructor guide.